Alexandre LOU-POUEYOU

M2 Research Internship Application: HPC Architecture & Performance Evaluation

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Profile

Master's student in Computer Science specializing in High Performance Computing and Data Science, with research experience in HPC I/O optimization and performance analysis. During my internship at Inria, I worked on the Gysela plasma simulation code and evaluated parallel I/O strategies on HPC systems. I am interested in parallel computing, distributed systems, data movement, and performance optimization.

Education

2024–present **Master's in Computer Science — HPC & Data Science**, *Expected Graduation: 2027*, University of Bordeaux / Enseirb, Talence
Relevant coursework: Parallel Architecture Programming (PAP), Network & Protocols, Distributed Systems, System Administration

2021–2024 **Bachelor's Degree in Computer Science**, *University of Bordeaux*, Talence
Strong foundation in algorithms, data structures, and low-level programming (C)

Experience

Nov 2026 – April 2027 **Part-time employee**, *Inria Center of the University of Bordeaux*, TADaaM Team, Talence
- **Large-Scale Benchmarking on Adastra:** Executed massive I/O performance experiments on CINES' Adastra supercomputer, scaling up to production-level configurations to project future exascale workloads.
- **Bayesian Optimization & Tuning:** Applied IOPS alongside Bayesian optimization strategies to systematically identify optimal netCDF/HDF5 and PDI parameter combinations for GYSELA.
- **CEA Collaboration & TOTO Integration:** Collaborated directly with the CEA Cadarache physics team to align optimization targets, while testing GYSELA integration with the in-house TOTO tool developed by TADaaM.
- **Scientific Writing:** Contributing to the drafting of a co-authored research paper documenting the optimization methodology and performance results.

Feb 2026 – July 2026 **HPC Internship**, *Inria Center of the University of Bordeaux*, TADaaM Team, Talence
- **IO & Checkpointing**: Optimized I/O performance and reduced checkpoints overhead in the Gysela plasma simulation code
- **Parallel Strategy & Overlap**: Evaluated parallel I/O strategies with PDI on BeegFS and Lustre, focusing on compute-I/O overlap to mitigate distributed bottlenecks
- **Benchmark & Profiling**: Conducted extensive performance profiling benchmarks on HPC clusters (Plafrim & Adastra) with IOPS tool.
- **Scientific Communication**: Presenting the progress made in my research during the internship at Compas2026 conference

2026–2027 **Network & System Administration**, *Association Label[i]*, Talence
- Administration of VPS infrastructure (deployment, monitoring, maintenance)
- Configuration and management of network services (web, reverse proxy, monitoring tools)
- Implementation of security practices and service isolation

2023–2025 **Seasonal Job — Payroll Management**, *Sodexo*, Bordeaux
Data processing, document management, and rigorous handling of administrative workflows

2022–2025 **President / Secretary**, *Association Label[i]*, Talence
Project management, team coordination, and organization of computer science events

References

Dr. Francieli Zanon Boito, *Associate Professor & Researcher*, Inria – TADaaM Team
Former Internship Supervisor (Feb. 2026 – July 2026).
Temporary Contract Supervisor (Nov 2026 - Apr 2027)
Email: francieli.zanon-boito@inria.fr
Recommendation letter: https://tinyurl.com/recommandation-francieli

Scientific Contributions

Ongoing work **Short Paper**, Scalability benchmark of DEISA-Dask for in-situ HPC analytics
Ivan Lucas, Ariel De Vora, Alexandre Lou-Poueyou, Hugo Strappazzon, Benoît Martin, Virginie Grandgirard.
Status: Ongoing work, intended for future submission.

Compas26 **Conference Presentation**, Improving I/O performances of the HPC Application Gysela
Alexandre Lou-Poueyou.
Presentation of resarch work at Compas2026, July 2026.

Academic & Personal Projects

Chess Engine (C++) Development of an optimized chess engine with a modular architecture. Focus on performance optimization, game tree search, and parallelization strategies (CPU/GPU).

Portfolio Website Personal portfolio to showcase projects and technical progress. Built with modern web technologies (Vite/Vue, SCSS).

Takuzu-2077 (C) Implementation of the Takuzu puzzle game with both console and graphical interfaces.

Skills

HPC & Parallel Computing MPI · OpenMP · CUDA · OpenCL · Slurm · Vectorization

HPC I/O & Storage PDI · Lustre · BeegFS · Parallel I/O · I/O Optimization

Programming C · C++ · Python · Cython · Java · TypeScript · Shell (Bash)

Systems & Tools Linux · Docker · Git · CI/CD · CMake · GDB · Valgrind · Flask

Core Interests Distributed Systems · Performance Profiling · Data-Intensive Applications

Languages

French Native

English Fluent (C1) – Linguaskill Certificate (Cambridge)

Japanese Beginner (A2)

Hobbies

Running (marathon), Strength training, Guitar, Reading, Chess